

Quick Start Guide

Installation (1 minute)

```
cd /home/ubuntu/altg
pip install -r requirements.txt
```

Setup API Keys (1 minute)

```
# Option 1: Environment variables
export OPENAI_API_KEY="your-key-here"
export ANTHROPIC_API_KEY="your-key-here"

# Option 2: Create .env file
cp .env.example .env
# Edit .env and add your keys
```

Run Your First Experiment (2 minutes)

```
# Quick test with GPT-3.5 (10 iterations)
python main.py \
  --name "quick_test" \
  --models "openai:gpt-3.5-turbo" \
  --strategies "random" \
  --iterations 10 \
  --batch-size 2
```

Results will be saved in `experiments/quick_test_<timestamp>/`

Run Examples (Interactive)

```
python example_usage.py
```

Choose from 6 interactive examples:

1. Basic Problem Generation
2. LLM Evaluation
3. Failure Analysis
4. Search Strategy
5. Visualization
6. Complete Experiment

Test Individual Modules

```
# Test problem generator (20 templates)
python problem_generator.py

# Test mutation engine (4 mutation types)
python mutation_engine.py

# Test evaluator
python evaluator.py

# Test visualizer
python visualizer.py
```

Common Commands

Single Model Test

```
python main.py --name "gpt4_test" --models "openai:gpt-4" --strategies "random" --iterations 50
```

Multi-Model Comparison

```
python main.py \
  --name "model_comparison" \
  --models "openai:gpt-4" "openai:gpt-3.5-turbo" "anthropic:claude-3-sonnet-20240229" \
  --strategies "random" "evolutionary" \
  --iterations 100
```

Full Experiment (All Strategies)

```
python main.py \
  --name "full_experiment" \
  --models "openai:gpt-4" \
  --strategies "random" "evolutionary" "hill_climbing" \
  --iterations 100 \
  --batch-size 5
```

Output Structure

```
experiments/
├── experiment_name_timestamp/
│   ├── results.json          # Metrics and analysis
│   ├── experiment.log        # Detailed logs
│   ├── logs/                 # Raw LLM responses
│   │   └── provider_model_responses.jsonl
│   └── visualizations/       # Charts and graphs
│       ├── failure_rates.png
│       ├── failure_rates.html
│       ├── mutation_dist_*.png
│       └── category_dist_*.png
```

Troubleshooting

API Key Not Found

```
# Verify keys are set
echo $OPENAI_API_KEY
echo $ANTHROPIC_API_KEY
```

Import Errors

```
# Reinstall dependencies
pip install -r requirements.txt --upgrade
```

Rate Limits

- Reduce `--batch-size` (try 1-2)
- Increase retry delay in `config.yaml`
- Use fewer iterations for testing

Next Steps

1. **Review Results:** Check `experiments/` for your experiment results
2. **View Visualizations:** Open HTML files in browser for interactive charts
3. **Analyze Failures:** Review `results.json` for detailed failure analysis
4. **Customize:** Edit `config.yaml` to adjust search parameters
5. **Read Full Docs:** See `README.md` for complete documentation

Tips

- Start with 10-20 iterations for quick tests
- Use `gpt-3.5-turbo` for rapid prototyping (cheaper/faster)
- Use `random` strategy first (simplest/fastest)
- Check logs/ for debugging LLM responses
- Interactive visualizations (HTML) allow zooming and hovering

Support

- Full documentation: `README.md`
- Example code: `example_usage.py`
- Module tests: Run any `.py` file directly
- Issues: Check logs in `experiments/*/experiment.log`

Happy testing! 🚀